

29.2 Mallows' Cp, AIC and BIC

A **selection criterion** assigns a number to each candidate model and picks the smallest. C_p estimates the prediction error of Section 29.1, AIC a likelihood-based discrepancy, and BIC the posterior probability of the model. The first two nearly always agree; the third answers a different question, and the difference shows up in large samples.

29.2.1 Candidates

The candidates are a finite collection of models $\mathcal{C}(\mathbf{X}_k)$, $k \in \mathcal{K}$, with $\text{rank}(\mathbf{X}_k) = p_k$, each fitted by least squares: $\hat{\boldsymbol{\mu}}_k = \mathbf{M}_k \mathbf{Y}$, with residual sum of squares $\text{SSE}_k = \|(\mathbf{I} - \mathbf{M}_k) \mathbf{Y}\|^2$. Typically $\mathbf{X}_k$ consists of the intercept and a subset of the columns of a full model matrix $\mathbf{X}_F$ of rank $P < n$, and $s^2 = \text{SSE}_F / (n - P)$ is the usual variance estimate from the full model. The data follow Eq. 29.1.1. By Theorem 29.1.1(c) the scaled risk of candidate k is

$$J_k = \frac{\mathbb{E} \|\hat{\boldsymbol{\mu}}_k - \boldsymbol{\mu}\|^2}{\sigma^2} = p_k + \gamma_k, \quad \gamma_k = \frac{\|(\mathbf{I} - \mathbf{M}_k) \boldsymbol{\mu}\|^2}{\sigma^2}, \quad (29.2.1)$$

variance plus squared bias. The bias term γ_k is the noncentrality Eq. 19.1.1 of the test of candidate k against any correct larger model.

29.2.2 Mallows' Cp

Definition 29.2.1. Mallows' C_p

The C_p statistic of candidate k is

$$C_k = \frac{\text{SSE}_k}{s^2} - n + 2p_k.$$

With σ^2 in place of s^2 , $\sigma^2(C_k + n)/n$ is the covariance-penalty estimate Eq. 29.1.3 of the in-sample prediction error of candidate k . The scaling makes C_k comparable with p_k .

Proposition 29.2.1. Properties of C_p

- (a) Under Eq. 29.1.1, $C_k^\sigma = \text{SSE}_k / \sigma^2 - n + 2p_k$ is unbiased for J_k .
- (b) Assume normal errors, $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X}_F)$, $\mathcal{C}(\mathbf{X}_k) \subseteq \mathcal{C}(\mathbf{X}_F)$ and $n - P > 2$. Then

$$\mathbb{E}(C_k) = J_k + \frac{2(P - p_k + \gamma_k)}{n - P - 2}.$$

- (c) Let $F_k = [(\text{SSE}_k - \text{SSE}_F) / (P - p_k)] / s^2$ be the F statistic for testing candidate k against the full model ($p_k < P$). Then

$$C_k = (P - p_k)(F_k - 1) + p_k.$$

In particular $C_F = P$, and $C_k \leq p_k$ iff $F_k \leq 1$.

Proof.

- (a) By Theorem 29.1.1(c), $E(\text{SSE}_k) = \sigma^2(n - p_k) + \|(\mathbf{I} - \mathbf{M}_k)\boldsymbol{\mu}\|^2 = \sigma^2(n - p_k + \gamma_k)$.
 (b) Write $\text{SSE}_k = \text{SSE}_F + (\text{SSE}_k - \text{SSE}_F)$. By Theorem 11.1.1(a), applied to $\mathcal{C}(\mathbf{X}_k) \subseteq \mathcal{C}(\mathbf{X}_F)$, the two terms are independent, $\text{SSE}_F/\sigma^2 \sim \chi^2(n - P)$, and $(\text{SSE}_k - \text{SSE}_F)/\sigma^2$ is noncentral χ^2 with $P - p_k$ degrees of freedom and noncentrality $\|(\mathbf{M}_F - \mathbf{M}_k)\boldsymbol{\mu}\|^2/\sigma^2 = \gamma_k$, so its mean is $P - p_k + \gamma_k$. By Lemma 14.1.3, $E(\sigma^2/s^2) = (n - P)/(n - P - 2)$. Hence

$$\begin{aligned} E\left(\frac{\text{SSE}_k}{s^2}\right) &= (n - P) + (P - p_k + \gamma_k) \frac{n - P}{n - P - 2} \\ &= n - p_k + \gamma_k + \frac{2(P - p_k + \gamma_k)}{n - P - 2}, \end{aligned}$$

and subtracting $n - 2p_k$ gives the result.

- (c) By the definition of F_k , $\text{SSE}_k/s^2 = (n - P) + (P - p_k)F_k$. Subtract $n - 2p_k$. ■

By (b), estimating σ^2 costs little unless the full model uses up most of the data. Mallows (1973) proposed plotting C_k against p_k : points near the line $C_p = p$ have small bias, and the lowest point estimates the best predictor. By (c), adding q columns changes C_p by $2q - (\text{SSE}_{\text{small}} - \text{SSE}_{\text{big}})/s^2$, so C_p prefers the larger model iff the F statistic of the added columns, with the full-model s^2 , exceeds 2.

29.2.3 AIC

If g is the true density and f a model density, the **Kullback–Leibler divergence** $\int g \log(g/f) \geq 0$ is zero iff $f = g$ (Kullback and Leibler 1951), and up to a term free of f it is $E_g[-\log f(\mathbf{Y}^*)]$ for a new data set $\mathbf{Y}^* \sim g$. A fitted model $f(\cdot; \hat{\boldsymbol{\theta}})$ is therefore judged by the **discrepancy**

$$\Delta = E_{\mathbf{Y}^*}[-2 \log f(\mathbf{Y}^*; \hat{\boldsymbol{\theta}}(\mathbf{Y}))],$$

with $\mathbf{Y}^*$ independent of $\mathbf{Y}$. The maximized log-likelihood evaluates the same function at the data themselves, and is optimistic in the same way as the training error.

Definition 29.2.2. AIC and BIC

For a model with d_k free parameters and maximized log-likelihood ℓ_k ,

$$\text{AIC}_k = -2\ell_k + 2d_k, \quad \text{BIC}_k = -2\ell_k + d_k \log n.$$

In the normal linear model with unknown variance, $d_k = p_k + 1$ and $-2\ell_k = n \log(2\pi \text{SSE}_k/n) + n$ (Theorem 7.3.1), so up to a constant common to all candidates,

$$\text{AIC}_k = n \log \frac{\text{SSE}_k}{n} + 2(p_k + 1), \quad \text{BIC}_k = n \log \frac{\text{SSE}_k}{n} + (p_k + 1) \log n. \quad (29.2.2)$$

Akaike (1973, 1974) showed that $2d$ is the leading term of the optimism of -2ℓ for a correct model fitted by maximum likelihood, in large samples. In the normal linear model the optimism can be computed exactly, without asymptotics.

Theorem 29.2.2. The expected discrepancy of a normal linear model

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$ with $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X}_k)$ and $n > p_k + 2$, and let $\hat{\sigma}_k^2 = \text{SSE}_k/n$. For the fitted density $N_n(\hat{\boldsymbol{\mu}}_k, \hat{\sigma}_k^2 \mathbf{I})$,

$$E(\Delta_k) = E[n \log(2\pi \hat{\sigma}_k^2)] + \frac{n(n + p_k)}{n - p_k - 2}.$$

Consequently

$$\text{AICc}_k = -2\ell_k + 2(p_k + 1) + \frac{2(p_k + 1)(p_k + 2)}{n - p_k - 2} \quad (29.2.3)$$

is an unbiased estimate of $E(\Delta_k)$.

Proof. For a new data set $\mathbf{Y}^* = \boldsymbol{\mu} + \boldsymbol{\varepsilon}^*$, $-2 \log f(\mathbf{Y}^*) = n \log(2\pi \hat{\sigma}_k^2) + \|\mathbf{Y}^* - \hat{\boldsymbol{\mu}}_k\|^2 / \hat{\sigma}_k^2$, and as in Eq. 29.1.2 $E(\|\mathbf{Y}^* - \hat{\boldsymbol{\mu}}_k\|^2 \mid \mathbf{Y}) = n\sigma^2 + \|\hat{\boldsymbol{\mu}}_k - \boldsymbol{\mu}\|^2$. So

$$\Delta_k = n \log(2\pi \hat{\sigma}_k^2) + \frac{n\sigma^2 + \|\mathbf{M}_k \boldsymbol{\varepsilon}\|^2}{\hat{\sigma}_k^2},$$

using $\hat{\boldsymbol{\mu}}_k - \boldsymbol{\mu} = \mathbf{M}_k \boldsymbol{\varepsilon}$ when $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X}_k)$. Now $\|\mathbf{M}_k \boldsymbol{\varepsilon}\|^2 / \sigma^2 \sim \chi^2(p_k)$ and $n\hat{\sigma}_k^2 / \sigma^2 = \|(\mathbf{I} - \mathbf{M}_k) \boldsymbol{\varepsilon}\|^2 / \sigma^2 \sim \chi^2(n - p_k)$ are independent (Theorem 7.5.1), and $E(\sigma^2 / \hat{\sigma}_k^2) = n / (n - p_k - 2)$ by Lemma 14.1.3. Hence the second term has mean $n(n + p_k) / (n - p_k - 2)$. For the second statement, $-2\ell_k = n \log(2\pi \hat{\sigma}_k^2) + n$, and the identity $n(n + p) = (n - p - 2)(n + 2(p + 1)) + 2(p + 1)(p + 2)$ shows that $n(n + p) / (n - p - 2) = n + 2(p + 1) + 2(p + 1)(p + 2) / (n - p - 2)$. ■

The corrected criterion AICc is due to Sugiura (1978) and Hurvich and Tsai (1989). Its extra term vanishes as $n \rightarrow \infty$ with p_k fixed, which recovers AIC. It matters when p_k is not small compared with n . With $n = 20$ and $p_k = 5$, AIC charges $2(p_k + 1) = 12$, while the exact charge in the proof is $2(p_k + 1)n / (n - p_k - 2) = 18.46$. A simulation of 100000 data sets gives a mean discrepancy of 68.11, equal to the theorem's value to two decimals. AIC underestimates it by 6.46, and the underestimate favours the larger models.

```
import numpy as np
rng = np.random.default_rng(2902)
n_c, p_c, reps_c = 20, 5, 100_000
Xc = np.column_stack([np.ones(n_c), rng.normal(size=(n_c, p_c - 1))])
mu_c = Xc @ np.linspace(1, 2, p_c) # the model is correct
H = Xc @ np.linalg.solve(Xc.T @ Xc, Xc.T)
Yc = mu_c + rng.normal(size=(reps_c, n_c))
fit_c = Yc @ H
s2ml = np.mean((Yc - fit_c) ** 2, axis=1) # ML variance estimate
# discrepancy: expected -2 log-likelihood of an independent copy Y*, at the fitted
# values
disc = n_c * np.log(2 * np.pi * s2ml) + (n_c + np.sum((fit_c - mu_c) ** 2, axis=1))
/ s2ml
target = np.mean(n_c * np.log(2 * np.pi * s2ml)) + n_c * (n_c + p_c) / (n_c - p_c -
2)
print(f"E(discrepancy) {disc.mean():.2f}, formula {target:.2f}")
print(f"penalty in AIC {2 * (p_c + 1)}, in AICc {2 * (p_c + 1) * n_c / (n_c - p_c -
2):.2f}")
```

[Run this code online](#)

For a wrong candidate the penalty is no longer free of μ , and $2d_k$ is an approximation, justified when the candidate is close to the truth (Konishi and Kitagawa 2008 discuss corrections). In practice AIC and C_p rank candidates almost identically ([Exercise \(B1\)](#)).

29.2.4 BIC

Suppose the candidates carry prior probabilities π_k , and the parameters θ_k of model k a prior density $\pi(\theta_k | k)$. The posterior probability of model k is proportional to $\pi_k m_k(\mathbf{y})$, where

$$m_k(\mathbf{y}) = \int f_k(\mathbf{y} | \theta_k) \pi(\theta_k | k) d\theta_k$$

is the **marginal likelihood** of the model. For the conjugate normal prior of Theorem 7.6.2 the integral has a closed form. BIC is an approximation that needs no particular prior.

Remark (Sketch: where $\log n$ comes from). This is a heuristic derivation; Schwarz (1978) proved the result for exponential families, and Kass and Raftery (1995) give conditions and error terms. Let $\hat{\theta}$ maximize $\ell(\theta) = \log f_k(\mathbf{y} | \theta)$, and let $\mathbf{J}_n = -\nabla^2 \ell(\hat{\theta})$ be the observed information, which for independent observations grows like n : $\mathbf{J}_n = n\tilde{\mathbf{J}}$ with $\tilde{\mathbf{J}}$ of order one. Expanding ℓ to second order about $\hat{\theta}$ (Laplace's method, Tierney and Kadane 1986),

$$\begin{aligned} m_k(\mathbf{y}) &\approx f_k(\mathbf{y} | \hat{\theta}) \pi(\hat{\theta} | k) \int e^{-(\theta - \hat{\theta})^\top \mathbf{J}_n (\theta - \hat{\theta})/2} d\theta \\ &= f_k(\mathbf{y} | \hat{\theta}) \pi(\hat{\theta} | k) (2\pi)^{d_k/2} \det(\mathbf{J}_n)^{-1/2}. \end{aligned}$$

Since $\log \det \mathbf{J}_n = d_k \log n + \log \det \tilde{\mathbf{J}}$,

$$-2 \log m_k(\mathbf{y}) = -2\ell_k + d_k \log n + O(1),$$

where the $O(1)$ collects the prior density at $\hat{\theta}$, $\det \tilde{\mathbf{J}}$ and 2π , none of which grows with n . So BIC_k approximates $-2 \log m_k$, and with equal prior probabilities the model with the smallest BIC is approximately the most probable.

The error is of order one, so $\exp(-\text{BIC}_k/2)$ is a crude estimate of a posterior probability, but the rate is right: each parameter costs $\log n$. For a “unit information” prior the error is of order $n^{-1/2}$ (Kass and Wasserman 1995).

29.2.5 Three criteria, one data set, one threshold

Adjusted R^2 of [Chapter 9](#) is sometimes used as a criterion too.

Proposition 29.2.3. Adjusted R^2 as a criterion

Maximizing $\bar{R}^2$ over candidates that contain the intercept is minimizing $s_k^2 = \text{SSE}_k/(n - p_k)$, and for two nested candidates the larger has the larger $\bar{R}^2$ iff its added columns have $F > 1$.

Proof. These are Proposition 9.5.3(d) and (c). ■

The threshold 1 is lower than any of the others, so $\bar{R}^2$ chooses the largest models. Since $E(s_k^2) = \sigma^2(1 + \gamma_k/(n - p_k))$ (Theorem 19.1.2(c)), minimizing s_k^2 aims at the smallest *bias*, not the smallest prediction error.

For two nested candidates, each criterion is a test with its own critical value. Let the larger model add q columns, have residual degrees of freedom ν , and let $F = [(SSE_{\text{small}} - SSE_{\text{big}})/q]/(SSE_{\text{big}}/\nu)$, so that $SSE_{\text{small}}/SSE_{\text{big}} = 1 + qF/\nu$. The larger model is preferred by

criterion	when	threshold for F , approximately
adjusted R^2	$F > 1$ (Proposition 29.2.3)	1
C_p	$F > 2$, with s^2 from the full model	2
AIC	$n \log(1 + qF/\nu) > 2q$	2
BIC	$n \log(1 + qF/\nu) > q \log n$	$\log n$
F test at level α	$F > F_\alpha(q, \nu)$	$F_\alpha(q, \nu)$

The approximations use $\log(1 + x) \approx x$ and $\nu \approx n$. With $n = 50$, $q = 1$ and $\nu = 47$, the exact thresholds are 1.92 for AIC and 3.82 for BIC ($\log n = 3.91$), against $F_{0.05}(1, 47) = 4.05$. As tests of one column, AIC acts at level 0.17 and BIC at level 0.056, close to the conventional 5% at this sample size. BIC's level falls as n grows, and AIC's does not.

Example. Violent crime in the states

Return to the 50 states of Chapter 6 (the District of Columbia left out) and take as response the logarithm of the violent crime rate in 2009. The candidate regressors are the percentages of adults with a high-school diploma, of people below the poverty line, of single-parent households, of white residents and of urban residents, so there are $2^5 = 32$ candidates, all with an intercept. The full model has $P = 6$ and $s^2 = 0.1035$. The eight candidates with the smallest C_p are

regressors	p	C_p	ΔAIC	ΔBIC	$\bar{R}^2$
single, urban	3	2.20	0.00	0.42	0.471
poverty, single, urban	4	2.47	0.09	2.42	0.480
single	2	3.54	1.49	0.00	0.445
high school, single, urban	4	3.94	1.72	4.05	0.463
high school, poverty, single, urban	5	4.01	1.57	5.82	0.474
single, white, urban	4	4.04	1.82	4.16	0.462

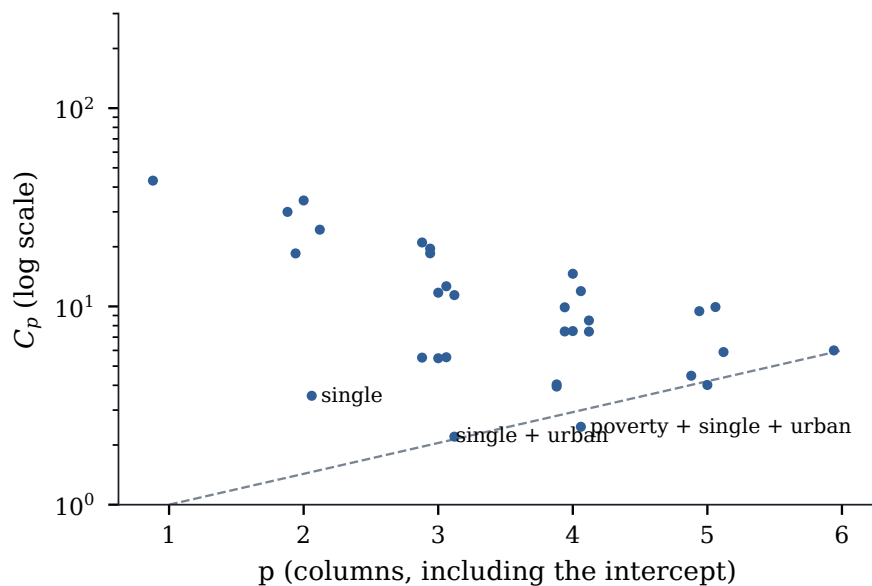


Figure 29.2.1: C_p against the number of columns for the 32 candidate models for log violent crime in the 50 states. The dashed line is $C_p = p$. The three labelled models are the choices of BIC, of C_p and AIC, and of adjusted R^2 .

poverty, single, white, urban	5	4.47	2.09	6.34	0.468
high school, single	3	5.48	3.42	3.85	0.434

where Δ is the difference from the smallest value over all 32 candidates. C_p and AIC choose single parenthood and urbanization; BIC drops urbanization; $\bar{R}^2$ adds poverty. The first three rows are within two units of each other on every scale, and none of the criteria separates them convincingly. Figure 29.2.1 is Mallows' plot. Every candidate without single parenthood lies far above the line $C_p = p$: it is badly biased.

```
import itertools
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statcrime.load_pandas().data.drop(index="District of Columbia")
names = ["hs_grad", "poverty", "single", "white", "urban"]
y = np.log(data["violent"].to_numpy())
n = len(y)

def sse(cols):
    X = np.column_stack([np.ones(n)] + [data[c].to_numpy() for c in cols])
    b, *_ = np.linalg.lstsq(X, y, rcond=None)
    return np.sum((y - X @ b) ** 2), X.shape[1]

sse_full, P = sse(names)
s2 = sse_full / (n - P)
# sigma^2 estimate from the
# largest model
sst = np.sum((y - y.mean()) ** 2)
```

```

rows = []
for k in range(len(names) + 1):
    for cols in itertools.combinations(names, k):
        e, p = sse(cols)
        rows.append(dict(model=cols, p=p, sse=e,
                        cp=e / s2 - n + 2 * p,
                        aic=n * np.log(e / n) + 2 * (p + 1),
                        bic=n * np.log(e / n) + np.log(n) * (p + 1),
                        adjr2=1 - (e / (n - p)) / (sst / (n - 1))))
for crit in ("cp", "aic", "bic"):
    best = min(rows, key=lambda r: r[crit])
    print(f"{crit:4s} chooses {best['model']}")
print("adj R2 chooses", max(rows, key=lambda r: r["adjr2"])["model"])

```

[Run this code online](#)

29.2.6 Consistency and efficiency

The thresholds explain the large-sample behaviour. The F statistic of an irrelevant column stays of order one, so a growing threshold such as $\log n$ eventually excludes it and a fixed one does not; that of a relevant column grows like n .

Proposition 29.2.4. AIC and BIC in large samples

Let the candidate set $\mathcal{K}$ be finite and fixed, and consider a sequence of designs and means with $n \rightarrow \infty$. Let the errors be independent and identically distributed with mean zero and variance σ^2 , and let $\sup_n \|\boldsymbol{\mu}\|^2/n < \infty$. Call k *correct* if $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X}_k)$. Assume that there is a correct k_0 such that every other correct candidate has $p_k > p_{k_0}$, and that $\liminf_n \|(\mathbf{I} - \mathbf{M}_k)\boldsymbol{\mu}\|^2/n > 0$ for every candidate that is not correct. Select $\hat{k}$ by minimizing $\text{IC}_k = n \log(\text{SSE}_k/n) + c_n p_k$.

- (a) If $c_n \rightarrow \infty$ and $c_n/n \rightarrow 0$, then $\Pr(\hat{k} = k_0) \rightarrow 1$. In particular BIC ($c_n = \log n$) is consistent.
- (b) If $c_n = c$ is constant, the errors are normal, and k is a correct candidate with $\mathcal{C}(\mathbf{X}_{k_0}) \subset \mathcal{C}(\mathbf{X}_k)$ and $q = p_k - p_{k_0}$, then $\Pr(\text{IC}_k < \text{IC}_{k_0}) \rightarrow \Pr\{\chi^2(q) > cq\} > 0$. In particular AIC ($c = 2$) is not consistent when such a candidate is available.
- (c) (*Efficiency, stated with a sketch.*) Suppose instead that no candidate is correct, and that the candidate sets grow with n so that the smallest risk over candidates tends to infinity. Then, under moment conditions on the errors and a condition limiting the number of candidates, with σ^2 known or consistently estimated in C_p , and for AIC with the largest candidate dimension $o(n)$, the loss $\|\hat{\boldsymbol{\mu}}_{\hat{k}} - \boldsymbol{\mu}\|^2$ of the C_p or AIC choice divided by $\min_k \|\hat{\boldsymbol{\mu}}_k - \boldsymbol{\mu}\|^2$ tends to one in probability (Shibata 1981; Li 1987). A criterion with $c_n \rightarrow \infty$ does not have this property in general.

Proof. Throughout, $\text{SSE}_k = \|(\mathbf{I} - \mathbf{M}_k)\boldsymbol{\mu}\|^2 + 2\boldsymbol{\mu}^\top(\mathbf{I} - \mathbf{M}_k)\boldsymbol{\varepsilon} + \|(\mathbf{I} - \mathbf{M}_k)\boldsymbol{\varepsilon}\|^2$. The last term is $\|\boldsymbol{\varepsilon}\|^2 - \|\mathbf{M}_k\boldsymbol{\varepsilon}\|^2$, where $\|\boldsymbol{\varepsilon}\|^2/n \rightarrow \sigma^2$ in probability by the weak law of large numbers and $\mathbb{E}\|\mathbf{M}_k\boldsymbol{\varepsilon}\|^2 = \sigma^2 p_k$, so $\|\mathbf{M}_k\boldsymbol{\varepsilon}\|^2 = O_p(1)$ by Markov's inequality. The middle term has mean zero and variance $4\sigma^2\|(\mathbf{I} - \mathbf{M}_k)\boldsymbol{\mu}\|^2 = O(n)$, so it is $O_p(\sqrt{n})$ by Chebyshev's inequality.

- (a) It suffices to show $\Pr(\text{IC}_k > \text{IC}_{k_0}) \rightarrow 1$ for each $k \neq k_0$, since $\mathcal{K}$ is finite. For k_0 , $\text{SSE}_{k_0} = \|\boldsymbol{\varepsilon}\|^2 - \|\mathbf{M}_{k_0}\boldsymbol{\varepsilon}\|^2$, so $\text{SSE}_{k_0}/n \rightarrow \sigma^2$.

A correct $k \neq k_0$. Then $|\text{SSE}_k - \text{SSE}_{k_0}| \leq \|\mathbf{M}_k \boldsymbol{\varepsilon}\|^2 + \|\mathbf{M}_{k_0} \boldsymbol{\varepsilon}\|^2 = O_p(1)$, and $n \log(\text{SSE}_k / \text{SSE}_{k_0}) = n \log(1 + (\text{SSE}_k - \text{SSE}_{k_0}) / \text{SSE}_{k_0}) = O_p(1)$, because the ratio inside the logarithm is $O_p(1/n)$. Hence $\text{IC}_k - \text{IC}_{k_0} = O_p(1) + c_n(p_k - p_{k_0})$, which tends to $+\infty$ in probability since $p_k > p_{k_0}$ and $c_n \rightarrow \infty$.

A candidate that is not correct. Let $0 < 2b < \liminf_n \|(\mathbf{I} - \mathbf{M}_k) \boldsymbol{\mu}\|^2 / n$. By the expansion, $\text{SSE}_k / n \geq \sigma^2 + b$ with probability tending to one, so $\log(\text{SSE}_k / \text{SSE}_{k_0}) \geq \delta$ with probability tending to one, where $\delta = \frac{1}{2} \log(1 + b/\sigma^2) > 0$. Then $\text{IC}_k - \text{IC}_{k_0} \geq n\delta - c_n|p_k - p_{k_0}|$, which tends to $+\infty$ because $c_n/n \rightarrow 0$.

- (b) Here $D = \text{SSE}_{k_0} - \text{SSE}_k = \|(\mathbf{M}_k - \mathbf{M}_{k_0}) \boldsymbol{\varepsilon}\|^2$, and $D/\sigma^2 \sim \chi^2(q)$ exactly for every n (Theorem 4.3.2). With $\text{SSE}_k/n \rightarrow \sigma^2$, Slutsky's theorem gives $nD/\text{SSE}_k \rightarrow \chi^2(q)$ in distribution, and $n \log(1 + D/\text{SSE}_k) - nD/\text{SSE}_k \rightarrow 0$ in probability because $D/\text{SSE}_k = O_p(1/n)$ and $|\log(1 + x) - x| \leq x^2$ for $x \geq 0$. So $n \log(\text{SSE}_{k_0}/\text{SSE}_k) \rightarrow \chi^2(q)$ in distribution. Now $\text{IC}_k < \text{IC}_{k_0}$ iff $n \log(\text{SSE}_{k_0}/\text{SSE}_k) > cq$, and cq is a continuity point of the limit.
- (c) *Sketch.* By Proposition 29.2.1(a), C_k^σ is unbiased for the scaled risk of each candidate, and $C_k^\sigma - J_k$ and the loss minus the risk are sums of centred quadratic and linear forms in $\boldsymbol{\varepsilon}$. The conditions make these fluctuations small compared with the risks, uniformly over the candidates, so minimizing C_k is asymptotically the same as minimizing the loss. Li (1987) gives such conditions for C_p , cross-validation and generalized cross-validation: finite moments of order $4m$ for the errors and $\sum_k (nR_k)^{-m} \rightarrow 0$ for some m , where R_k is the risk of candidate k per observation. AIC differs from C_p by terms of order p_k^2/n , which are negligible when the candidate dimensions are $o(n)$ (Shibata 1981). A growing penalty c_n biases the choice towards candidates whose risk is larger by a factor that does not tend to one, as Figure 29.2.2(b) shows. ■

For AIC the limits in (b) are 0.157, 0.135 and 0.075 for $q = 1, 2, 5$: one extra column is kept about one time in six however large the sample. Nishii (1984) proved (a) and (b) for the normal linear model.

Example. Consistency against efficiency

Panel (a) of Figure 29.2.2 uses six independent normal regressors, three of them with slopes 1, 0.5 and 0.25 and three with slope zero, and all 64 subsets as candidates. With $n = 50$ the true subset is found by AIC with probability 0.28 and by BIC with probability 0.24, because the smallest slope is hard to detect. With $n = 3200$ BIC finds it with probability 0.99, while AIC stays at 0.60. That is the limit $\Pr\{\chi^2(1) \leq 2\}^3 = 0.598$, because in large samples AIC keeps each irrelevant regressor independently when its squared t statistic exceeds about 2.

In panel (b) the truth has 1000 regressors with slopes $2/j$, all nonzero, and the candidates are the nested models containing the first k regressors, $k \leq n/4$. No candidate is correct. The ratio of the loss of the chosen model to the smallest loss among the candidates falls for AIC, from 1.18 at $n = 100$ to 1.07 at $n = 3200$, while for BIC it stays between 1.3 and 1.8 at every sample size.

```
import itertools
import numpy as np
rng = np.random.default_rng(2903)
beta = np.array([1.0, 0.5, 0.25, 0.0, 0.0, 0.0]) # three relevant, three
            irrelevant
```

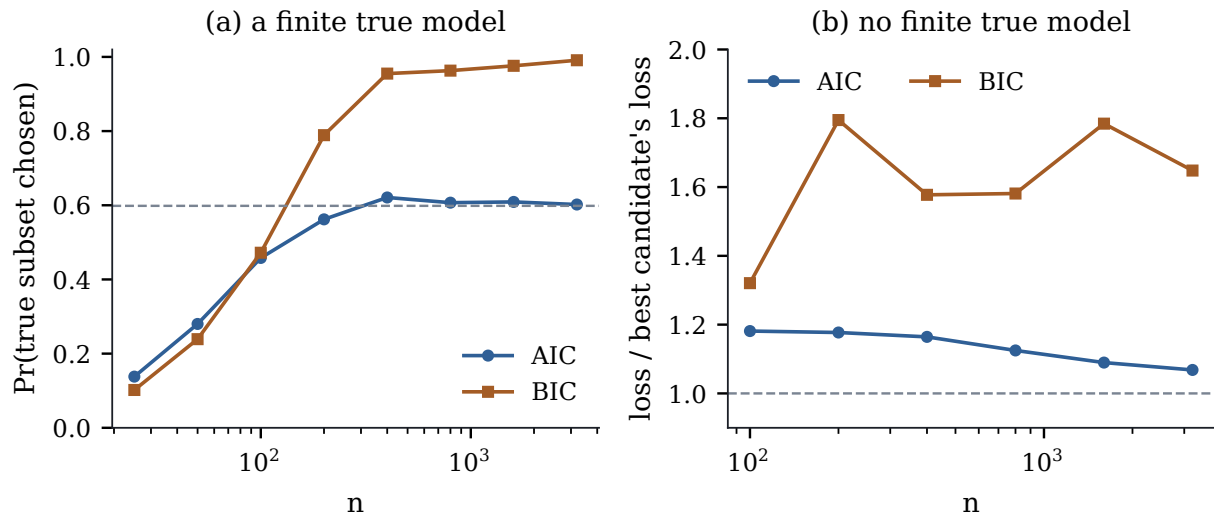


Figure 29.2.2: (a) Probability of selecting the true subset among 64, when a finite true model exists; dashed: AIC's limit 0.598. (b) Loss of the selected model relative to the best candidate, when no candidate is true. 1000 simulated data sets per point in (a), 400 in (b).

```

true_set = (0, 1, 2)
subsets = [s for k in range(7) for s in itertools.combinations(range(6), k)]

def choose(X, y, penalty):
    """All-subsets choice by  $n \log(SSE/n) + \text{penalty} * (\text{number of columns})$ ."""
    n = len(y)
    best, arg = np.inf, None
    for s in subsets:
        Xs = np.column_stack([np.ones(n), X[:, list(s)]])
        b, *_ = np.linalg.lstsq(Xs, y, rcond=None)
        crit = n * np.log(np.sum((y - Xs @ b) ** 2) / n) + penalty * Xs.shape[1]
        if crit < best:
            best, arg = crit, s
    return arg

def hit_rates(n, reps):
    X = rng.normal(size=(n, 6)) # the design, fixed for this n
    hits = np.zeros(2)
    for _ in range(reps):
        y = 1 + X @ beta + rng.normal(size=n)
        hits += [choose(X, y, 2.0) == true_set, choose(X, y, np.log(n)) == true_set]
    return hits / reps

for n in (50, 200, 800):
    aic, bic = hit_rates(n, 100)
    print(f"n = {n:4d}: AIC finds the true subset {aic:.2f} of the time, BIC {bic:.2f}")

```

[Run this code online](#)

The two properties cannot be combined: Yang (2005) showed that no rule can be consistent and

also attain the best rate of prediction risk. The choice between AIC and BIC is a choice of goal: prediction (AIC, C_p , cross-validation) or identifying a small true set of regressors (BIC).

29.2.7 Selection in an orthonormal design

When the columns are orthonormal the search disappears, and the criteria can be compared exactly.

Proposition 29.2.5. Selection as hard thresholding

Let $\mathbf{X}$ have orthonormal columns, $\mathbf{X}^\top \mathbf{X} = \mathbf{I}_m$, put $\mathbf{Z} = \mathbf{X}^\top \mathbf{Y}$, and let σ^2 be known. Among all subsets S of the columns, $\text{SSE}_S/\sigma^2 + c^2|S|$ is minimized (uniquely, with probability one) by $S = \{j : |Z_j| > c\sigma\}$. The selected fit has coefficients $Z_j 1\{|Z_j| > c\sigma\}$.

Proof. The projection onto the columns in S is $\sum_{j \in S} \mathbf{x}_j \mathbf{x}_j^\top$, so $\text{SSE}_S = \|\mathbf{Y}\|^2 - \sum_{j \in S} Z_j^2$ and the criterion is $\|\mathbf{Y}\|^2/\sigma^2 + \sum_{j \in S} (c^2 - Z_j^2/\sigma^2)$. A column lowers it iff $Z_j^2 > c^2\sigma^2$, and ties have probability zero. ■

With known σ^2 , C_p and AIC take $c = \sqrt{2}$ and BIC takes $c = \sqrt{\log n}$. Testing each coefficient at level 5% and keeping the significant ones (the **pretest estimator** of Bancroft 1944) takes $c = 1.96$. This is the hard thresholding of Theorem 27.4.2, and each coordinate can be studied alone. With $\sigma = 1$ and $Z \sim N(\theta, 1)$, the risk of $Z 1\{|Z| > c\}$ is

$$\begin{aligned} R(\theta; c) &= \theta^2 D + 1 - D + (c - \theta)\phi(c - \theta) + (c + \theta)\phi(c + \theta), \\ D &= \Phi(c - \theta) - \Phi(-c - \theta), \end{aligned} \tag{29.2.4}$$

where D is the probability that the coordinate is dropped ([Exercise \(B2\)](#)). Least squares has risk 1, and dropping the coordinate always has risk θ^2 .

[Figure 29.2.3](#) shows the trade. At $\theta = 0$ the risks are 0.572 for $c = \sqrt{2}$, 0.279 for the pretest and 0.075 for BIC with $n = 1000$. (At zero the risk equals the degrees of freedom of Proposition 29.1.3: both are $E[Z^2 1\{|Z| > c\}]$.) The curves cross 1 at $\theta = 0.78, 0.84$ and 0.92 and peak at 1.651, 2.464 and 3.956, at $\theta = 1.88, 2.16$ and 2.59 . A larger threshold gains more at zero and loses more at moderate θ , and no threshold, least squares included, is uniformly best. This is Theorem 19.1.5 again: dropping a coordinate helps iff $\theta^2 < 1$, and the data cannot tell which side of the boundary they are on. Shrinkage avoids the bump: the James–Stein estimator of Theorem 27.5.2 beats least squares everywhere when $m \geq 3$.

```
import numpy as np
from scipy.stats import norm

def hard_risk(theta, c):
    """E(Z 1{|Z| > c} - theta)^2 for Z ~ N(theta, 1)."""
    D = norm.cdf(c - theta) - norm.cdf(-c - theta) # Pr(|Z| <= c): coordinate
    # dropped
    return theta**2 * D + 1 - D + (c - theta) * norm.pdf(c - theta) + (c + theta) *
    norm.pdf(c + theta)
```

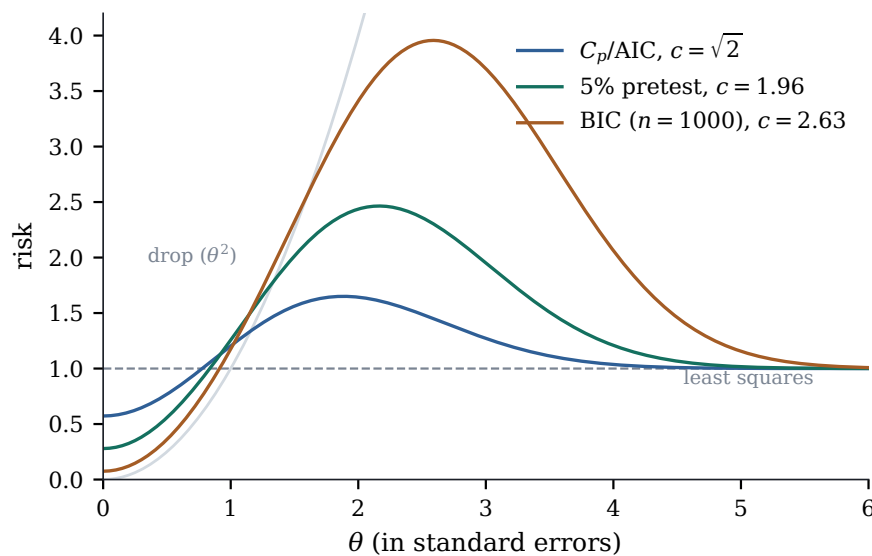


Figure 29.2.3: Risk of hard thresholding $Z1\{|Z| > c\}$, $Z \sim N(\theta, 1)$, for the thresholds of C_p and AIC, a 5% pretest and BIC with $n = 1000$. Least squares has risk 1 (dashed); always dropping the coordinate has risk θ^2 (grey).

```
thetas = np.linspace(0, 6, 601)
for label, c in [("C_p / AIC", np.sqrt(2)), ("5% pretest", 1.96), ("BIC, n = 1000",
    np.sqrt(np.log(1000)))]:
    r = hard_risk(thetas, c)
    print(f"{label:14s} c = {c:.3f}: risk at 0 = {r[0]:.3f}, "
        f"maximum {r.max():.3f} at theta = {thetas[r.argmax():.2f}]")
```

[Run this code online](#)

29.2.8 Exercises

29.2.8.1 A. Check your understanding

Exercise. A1

In [Example \(Violent crime in the states\)](#) the model with single parenthood and urbanization has $SSE = 4.782$. Compute its C_p from s^2 , n and p , and the F statistic for testing it against the full model.

Solution. $C_p = 4.782/0.1035 - 50 + 6 = 46.20 - 44 = 2.20$. By [Proposition 29.2.1\(c\)](#), $2.20 = 3(F - 1) + 3$, so $F = 0.73$: the three regressors left out together have an F statistic below one.

29.2.8.2 B. Practice

Exercise. B1

Let $\hat{\sigma}^2$ be any fixed positive number and write $\text{SSE}_k/n = \hat{\sigma}^2(1 + \delta_k)$. Show that

$$\text{AIC}_k - \text{AIC}_j = \frac{\text{SSE}_k - \text{SSE}_j}{\hat{\sigma}^2} + 2(p_k - p_j) + n O(\delta_k^2 + \delta_j^2),$$

and conclude that with $\hat{\sigma}^2 = s^2$ AIC and C_p differences agree when every candidate's residual mean square is close to s^2 .

Exercise. B2

Derive [Eq. 29.2.4](#). (Split the risk according to whether the coordinate is kept, and use $\int_a^b w^2 \phi(w) dw = \Phi(b) - \Phi(a) - b\phi(b) + a\phi(a)$.)

Solution. Let $Z = \theta + W$. When the coordinate is dropped ($|Z| \leq c$, probability D) the loss is θ^2 ; when kept it is W^2 . So $R = \theta^2 D + E[W^2 1\{|\theta + W| > c\}]$. The event $|\theta + W| \leq c$ is $-c - \theta \leq W \leq c - \theta$, so $E[W^2 1\{|\theta + W| > c\}] = 1 - \int_{-c-\theta}^{c-\theta} w^2 \phi(w) dw = 1 - D + (c - \theta)\phi(c - \theta) + (c + \theta)\phi(c + \theta)$, using $\phi(-x) = \phi(x)$.

Exercise. B3

Show that AIC prefers the larger of two nested candidates iff $F > (\nu/q)(e^{2q/n} - 1)$, and BIC iff $F > (\nu/q)(n^{q/n} - 1)$. Find the limits of both thresholds as $n \rightarrow \infty$ with q and $n - \nu$ fixed.

29.2.8.3 C. Going deeper

Exercise. C1

In panel (a) of [Example \(Consistency against efficiency\)](#), suppose the three regressors with zero slopes are orthogonal to each other and to the relevant ones. Show that for large n AIC selects the true subset iff each of them has squared t statistic below about 2, and deduce the limit $\Pr\{\chi^2(1) \leq 2\}^3$.

Solution. Every candidate that omits a relevant regressor loses with probability tending to one ([Proposition 29.2.4](#), proof of (a), second case, which holds for any c_n with $c_n/n \rightarrow 0$). Among candidates containing the relevant block, orthogonality makes $\text{SSE}_{k_0 \cup A} = \text{SSE}_{k_0} - \sum_{j \in A} \|\mathbf{M}_{\{j\}} \boldsymbol{\epsilon}\|^2$, and $n \log(\text{SSE}_{k_0}/\text{SSE}_{k_0 \cup A}) = \sum_{j \in A} T_j^2 + o_p(1)$ with $T_j^2 \rightarrow \chi^2(1)$ independent. AIC adds A iff this exceeds $2|A|$, so it chooses $A = \{j : T_j^2 > 2\}$ up to $o_p(1)$, and chooses $A = \emptyset$ with probability tending to $\prod_j \Pr(T_j^2 \leq 2) = \Pr\{\chi^2(1) \leq 2\}^3$.